

Comprehensive Mathematical & Structural Engine Formulae (CivilApp)

This is the extended, exhaustive reference of every physical, geometrical, and structural equation utilized in the software from start to finish.

1. DXF Geometry Processing & Auto-Framing

1.1 Wall Centerline Normalization

When double lines (representing inner and outer wall faces) are detected, the structural centerline is computed as:

$$X_{center} = (X_{inner} + X_{outer}) / 2$$

$$Y_{center} = (Y_{inner} + Y_{outer}) / 2$$

1.2 Structural Grid Generation & Snapping

Nodes are merged if their distance 'd' is less than the snapping tolerance (usually 150mm):

$$d = \sqrt{(X_2 - X_1)^2 + (Y_2 - Y_1)^2} \leq 150 \text{ mm}$$

1.3 Span Limitation Check

To prevent uneconomical beam depths, spans are mathematically restricted:

$$\text{If } L_{span} > 5000 \text{ mm} \rightarrow \text{Insert intermediate column at } L/2$$

2. Load Takedown & Gravity Calculations

2.1 Unit Weights (IS 875 Part 1)

- Reinforced Concrete: 25 kN/m³
- Burnt Clay Brick: 19.2 kN/m³
- AAC Block: 7.5 kN/m³
- Steel: 7850 kg/m³

2.2 Tributary Area Method (Slab Loads)

For a grid column at intersection (i, j):

$$A_{trib} = [(L_{x_left} + L_{x_right})/2] * [(L_{y_top} + L_{y_bottom})/2]$$

$$P_{slab} = A_{trib} * (Dead_Load + Live_Load)$$

2.3 Masonry Wall Loads

For a beam supporting a wall of height H_{story} and thickness t_{wall} :

$$w_{wall} = \text{Density}_{masonry} * t_{wall} * (H_{story} - D_{beam}) \text{ [kN/m]}$$

The nodal point load transferred to columns (half to each end):

$$P_{wall_transfer} = (w_{wall} * L_{span}) / 2$$

2.4 Load Combinations (IS 456 Table 18)

- Gravity Limit State: $1.5 * (DL + LL)$
- Seismic Limit State: $1.2 * (DL + LL \pm EQ)$ or $1.5 * (DL \pm EQ)$

3. Finite Element Analysis (FEA) Engine

3.1 Element Local Stiffness Matrix [k]

For a 2D frame beam element (axial, shear, moment per node), a 6x6 stiffness matrix is generated:

```
[k] =  
[ EA/L 0 0 -EA/L 0 0 ]  
[ 0 12EI/L^3 6EI/L^2 0 -12EI/L^3 6EI/L^2 ]  
[ 0 6EI/L^2 4EI/L 0 -6EI/L^2 2EI/L ]  
[ -EA/L 0 0 EA/L 0 0 ]  
[ 0 -12EI/L^3 -6EI/L^2 0 12EI/L^3 -6EI/L^2 ]  
[ 0 6EI/L^2 2EI/L 0 -6EI/L^2 4EI/L ]
```

3.2 Fixed End Moments (FEM)

For a Uniformly Distributed Load 'w' acting on the beam:

```
M_FEM_A = (w * L^2) / 12  
M_FEM_B = -(w * L^2) / 12  
V_FEM = (w * L) / 2
```

3.3 Global System Solution

Transformation matrix [T] rotates [k] to global axes:

```
[K_global] = [T]^T * [k] * [T]
```

The system solves for nodal displacements {u} using the inverse stiffness matrix:

```
{F} = [K_global] * {u} --> {u} = [K_global]^-1 * {F}
```

4. Column Design (IS 456:2000)

4.1 Slenderness Check (Cl. 25.1.2)

```
Slenderness Ratio = L_eff / b_min  
If Ratio < 12, it is a Short Column.
```

4.2 Minimum Eccentricity (Cl. 25.4)

```
e_min = L/500 + D/30 (Must be >= 20 mm)
```

4.3 Axial Capacity & Steel Sizing (Cl. 39.3)

Gross concrete area sizing (Before steel):

```
P_u <= 0.4 * fck * Ag --> Ag_req = P_u / (0.4 * fck)  
Solving for longitudinal reinforcement area (Asc):  
P_u = 0.4 * fck * (Ag - Asc) + 0.67 * fy * Asc  
Asc = (P_u - 0.4 * fck * Ag) / (0.67 * fy - 0.4 * fck)
```

Limits: 0.8% Ag <= Asc_provided <= 4.0% Ag

5. Beam Design (IS 456 Annex G)

5.1 Effective Depth Heuristics

Using span-to-depth ratios to prevent excessive deflection (CI 23.2.1):

$$D_{req} = L_{span} / 12 \text{ (Conservative assumption over standard 20)}$$

5.2 Ultimate Moment Capacity & Neutral Axis

Depth of neutral axis (x_u):

$$x_u = (0.87 * f_y * A_{st}) / (0.36 * f_{ck} * b)$$

Limiting neutral axis depth for Fe415 steel:

$$x_{u_{max}} = 0.48 * d$$

Moment of Resistance (M_u):

$$M_u = 0.87 * f_y * A_{st} * d * [1 - (A_{st} * f_y) / (b * d * f_{ck})]$$

5.3 Shear Design (Stirrups)

Nominal shear stress:

$$\tau_{u_v} = V_u / (b * d)$$

Design shear strength of concrete (τ_{u_c}) is interpolated based on:

$$p_t = (100 * A_{st}) / (b * d)$$

Required shear force for stirrups:

$$V_{us} = V_u - (\tau_{u_c} * b * d)$$

Stirrup spacing S_v :

$$S_v = (0.87 * f_y * A_{sv} * d) / V_{us} \text{ } (<= 0.75d \text{ or } 300\text{mm})$$

6. Slab Design Basics

6.1 Thickness Calculation

$$d_{slab} = L_{short} / 26 \text{ (Continuous)}$$

$$d_{slab} = L_{short} / 20 \text{ (Simply Supported)}$$

Minimum overall depth = 125 mm or 150 mm

6.2 Minimum Distribution Steel

For High Yield Strength Deformed (HYSD) bars (Fe415/Fe500):

$$A_{st_{min}} = 0.12\% * b * D$$

7. Isolated Footing Design

7.1 Base Area Sizing

Adding 10% self-weight of the footing:

$$P_{design} = P_{axial} * 1.1$$

$$A_{req} = P_{design} / SBC$$

$$\text{Side of square footing } B = \sqrt{A_{req}}$$

7.2 Punching Shear (Two-Way Shear)

Critical perimeter is at a distance $d/2$ from column face.

Allowable shear stress:

$$\tau_{u_c} = 0.25 * \sqrt{f_{ck}} \text{ [in N/mm}^2\text{]}$$

7.3 Flexure (Bending)

Moment at the face of the column:

$$M_u = (p_{net} * L_{cantilever}^2 / 2) * B$$

8. Lateral Dynamics: Earthquake (IS 1893:2016)

8.1 Seismic Weight Calculation

Live Load reduction based on Table 10:

$$W = \text{Sum}(\text{DL}) + \text{Sum}(\text{Fraction} * \text{LL}) \text{ (Roof LL is always 0\%)}$$

8.2 Fundamental Time Period

For buildings with masonry infill panels:

$$T_a = (0.09 * h) / \text{sqrt}(d)$$

8.3 Design Base Shear

Horizontal Seismic Coefficient (A_h):

$$A_h = (Z / 2) * (I / R) * (S_a / g)$$

Total Base Shear:

$$V_B = A_h * W$$

9. Lateral Dynamics: Wind (IS 875 Part 3)

9.1 Design Wind Speed

$$V_z = V_b * k_1 * k_2 * k_3 * k_4$$

9.2 Wind Pressure & Force

$$P_z = 0.6 * V_z^2 \text{ [N/m}^2\text{]}$$

Total story force:

$$F = P_z * (\text{Width} * \text{Story_Height})$$

10. Bar Bending Schedule (BBS) Metrics

10.1 Development Length (L_d)

For deformed bars in tension (IS 456 Cl. 26.2.1):

$$L_d = (\phi * 0.87 * f_y) / (4 * \tau_{bd})$$

10.2 Cutting Length of Box Stirrups

For a column of size A x B with clear cover c:

$$x = A - 2c, \quad y = B - 2c$$

$$L_{cut} = 2*(x + y) + 24*\phi \text{ (Includes hook length)}$$

11. Bill of Materials (BOM) & Cost Calculations

11.1 Masonry Volume to Unit Conversion

Number of standard Indian bricks (190x90x90 mm) with 10mm mortar:

$$\text{Volume per brick} = 0.2 * 0.1 * 0.1 = 0.002 \text{ m}^3$$

$$\text{Bricks per m}^3 = 1 / 0.002 = 500 \text{ bricks}$$

11.2 Mortar & Plaster Calculations

Mortar occupies roughly 30% of total brickwork volume:

$$V_{\text{mortar}} = V_{\text{wall_total}} * 0.30$$

Plastering Surface Area (Double-sided):

$$A_{\text{plaster}} = (L_{\text{wall}} * H_{\text{wall}} - A_{\text{openings}}) * 2$$

11.3 Total Structural Cost Aggregation

$$\text{Cost} = (V_{\text{conc}} * 6000) + (W_{\text{steel}} * 70) + (\text{Bricks} * 8) + (V_{\text{excav}} * 250)$$